

Momentum-Space QCD Sum-Rule Setup for the $B_c(1P)$ Axial-Vector Mixing Angle

1 Purpose and conventions

We study the mixing between the two axial-vector $B_c(1P)$ basis states associated with the 3P_1 and 1P_1 structures. The current implementation follows the Aliev et al. mixing-angle method, but the OPE is evaluated in momentum space because both quark lines are heavy.

This note records the calculation implemented in `BcMixingMomentum.wl`. It is written so that every convention and intermediate object can be checked independently before using the numerical angle in a paper.

The quark content convention is

$$B_c^+ = \bar{b}c. \quad (1)$$

The two interpolating currents are taken as

$$J_\mu^A = \bar{b} \gamma_\mu \gamma_5 c, \quad (2)$$

$$J_\mu^B = i \bar{b} \sigma_{\mu\alpha} \frac{p^\alpha}{m_b + m_c} \gamma_5 c, \quad (3)$$

where

$$\sigma_{\mu\nu} = \frac{i}{2} [\gamma_\mu, \gamma_\nu]. \quad (4)$$

The explicit factor of i in J_μ^B is retained. This is important because `FeynCalc`'s `DiracSigma` already contains the conventional $i/2$ commutator.

The factor $1/(m_b + m_c)$ is also essential. Since $[\bar{Q}Q] = 3$ and $[p^\alpha] = 1$, the unnormalized tensor current $i\bar{b}\sigma_{\mu\alpha}p^\alpha\gamma_5c$ has mass dimension 4, while J_μ^A has mass dimension 3. The normalized current above makes both currents dimension 3.

If the opposite flavor convention is used, for example $B_c^- = \bar{c}b$, the sign of the quoted angle can change. The absolute convention must therefore be kept fixed when comparing to other papers.

2 Mass-dimension check

The quark field has dimension $3/2$, while p^α , m_b , and m_c have dimension 1. Therefore

$$[J_\mu^A] = 3, \quad [J_\mu^B] = 3 \quad (5)$$

only after the tensor current is normalized by $m_b + m_c$.

For currents with dimensions d_i and d_j ,

$$[\Pi_{\mu\nu}^{ij}] = d_i + d_j - 4, \quad (6)$$

because of the d^4x integration in the two-point function. Thus the normalized basis gives

$$[\Pi_1^{AA}] = [\Pi_1^{AB}] = [\Pi_1^{BB}] = 2. \quad (7)$$

The spectral densities have the same mass dimension as the invariant correlator,

$$[\rho^{AA}] = [\rho^{AB}] = [\rho^{BB}] = 2. \quad (8)$$

In the present code the Borel moment is evaluated as $\int ds e^{-s/M^2} \rho(s)$, so all three numerical moments carry one common extra factor of $[ds] = 2$. This common convention does not affect the mixing angle.

If the tensor current were left unnormalized, the dimensions would instead be

$$[\rho^{AA}], [\rho^{AB}], [\rho^{BB}] = 2, 3, 4, \quad (9)$$

and the matrix entries in the angle formula would not be dimensionally compatible.

3 Implementation map

The main code objects are:

- `Correlator[channel, order]`: $\Pi_1^{ij}(p^2)$ after FeynCalc Dirac algebra and spin projection.
- `SpectralDensity[channel, ‘‘pert’’, s]`: $\rho_{\text{pert}}^{ij}(s)$.
- `NumericBorelPi[channel, order, M2, s0]`: $\Pi_{\text{order}}^{ij}(M^2, s_0)$.
- `NumericMixingAngleDegrees[M2, s0, ‘‘total’’]`: $\theta(M^2, s_0)$ from $\Pi_{\text{pert}}^{ij} + \Pi_{G^2}^{ij} + \Pi_{G^3}^{ij}$.

The implemented OPE orders are

$$\text{pert}, \quad G^2, \quad \text{pertG2} = \text{pert} + G^2, \quad G^3, \quad \text{total} = \text{pert} + G^2 + G^3. \quad (10)$$

The implemented dimension-6 triple-gluon condensate $\langle g_s^3 G^3 \rangle$ is the standard vacuum-averaged single-line propagator term. Possible cross-line open-field G^3 terms need a separate derivation.

4 How to reproduce the calculation while reading this note

This section is a practical roadmap for a new student. The goal is not to evaluate the final number immediately, but to check one layer at a time. A useful order is:

1. Check the current definitions and mass dimensions. In particular, verify that the factor $p^\alpha/(m_b + m_c)$ in J_μ^B makes the AA , AB , and BB correlators dimensionally compatible.
2. Reproduce the Wick contractions for $\Pi_{\mu\nu}^{AA}$, $\Pi_{\mu\nu}^{AB}$, and $\Pi_{\mu\nu}^{BB}$. At this stage do not calculate any integrals; only check the order of gamma matrices and signs.
3. Apply the spin-1 projector

$$P_{(1)}^{\mu\nu} = \frac{1}{3} \left(g^{\mu\nu} - \frac{p^\mu p^\nu}{p^2} \right)$$

and verify that a scalar invariant Π_1^{ij} is obtained.

4. For the perturbative term, put the two internal quarks on shell and verify the three numerator functions N^{AA} , N^{AB} , and N^{BB} given below. This is the most important algebraic check.
5. Verify the phase-space factor

$$\frac{N_c}{16\pi^2} \frac{\sqrt{\lambda(s, m_b, m_c)}}{s}$$

and the physical threshold $s_{\text{th}} = (m_b + m_c)^2$.

6. Perform the Borel integral

$$\Pi^{ij}(M^2, s_0) = \int_{s_{\text{th}}}^{s_0} ds e^{-s/M^2} \rho^{ij}(s)$$

and only then insert the three Borel moments into the mixing-angle formula.

7. Add condensates only after the perturbative result is stable. For G^2 , keep the local single-line terms and the cross-line open-field term conceptually separate; otherwise it is easy to double count or miss a contribution.

The Mathematica workflow mirrors these steps:

Hand calculation object	Mathematica check
current dimensions	<code>MixingMassDimensionReport[]</code>
projected trace	<code>Correlator[channel, "pert"]</code>
perturbative numerator	<code>PerturbativeNumerator[channel, s]</code>
spectral density	<code>PerturbativeSpectralDensity[channel, s]</code>
Borel moment	<code>NumericBorelPi[channel, order, M2, s0]</code>
mixing angle	<code>NumericMixingAngleDegrees[M2, s0, order]</code>
coordinate trace	<code>CoordinateTraceKernel[channel, order]</code>
coordinate angle	<code>CoordinateNumericMixingAngleDegrees[M2, s0, order]</code>

When a new term is added, the recommended debugging procedure is:

1. compute one channel, usually AA , at one point;
2. check that the result is real;
3. check its mass dimension;
4. check its sign and size against the corresponding momentum-space expression, if available;
5. only then scan over M^2 and s_0 .

5 Correlators and spin-1 projection

Define

$$\Pi_{\mu\nu}^{ij}(p) = i \int d^4x e^{ip \cdot x} \langle 0 | T \{ J_\mu^i(x) J_\nu^{j\dagger}(0) \} | 0 \rangle, \quad i, j \in \{A, B\}. \quad (11)$$

The invariant amplitude used for the axial-vector spin-1 state is extracted with

$$P_{(1)}^{\mu\nu} = \frac{1}{3} \left(g^{\mu\nu} - \frac{p^\mu p^\nu}{p^2} \right), \quad (12)$$

so that

$$\Pi_1^{ij}(p^2) = P_{(1)}^{\mu\nu} \Pi_{\mu\nu}^{ij}(p). \quad (13)$$

The mixing angle is then obtained from

$$\tan 2\theta = -\frac{2\Pi_1^{AB}}{\Pi_1^{AA} - \Pi_1^{BB}}. \quad (14)$$

Numerically we use the quadrant-safe definition

$$\theta_{\text{raw}} = \frac{1}{2} \text{atan2} \left(-2\Pi_1^{AB}, \Pi_1^{AA} - \Pi_1^{BB} \right), \quad (15)$$

and quote the principal branch

$$\theta = \theta_{\text{raw}} - \frac{\pi}{2} \text{Round} \left(\frac{\theta_{\text{raw}}}{\pi/2} \right), \quad \theta \in [-45^\circ, 45^\circ]. \quad (16)$$

6 Momentum-space OPE

The perturbative heavy-quark propagator is

$$S_Q^{(0)}(k) = \frac{\not{k} + m_Q}{k^2 - m_Q^2}. \quad (17)$$

For the dimension-4 gluon condensate we use the background-field expansion

$$S_Q(k) = S_Q^{(0)}(k) + g_s G_A^{\alpha\beta} T^A S_{Q,\alpha\beta}^{(G)}(k) + S_Q^{(G^2)}(k), \quad (18)$$

where

$$S_{Q,\alpha\beta}^{(G)}(k) = -\frac{1}{4} \frac{\sigma_{\alpha\beta}(\not{k} + m_Q) + (\not{k} + m_Q)\sigma_{\alpha\beta}}{(k^2 - m_Q^2)^2}, \quad (19)$$

$$S_Q^{(G^2)}(k) = \frac{\langle g_s^2 G^2 \rangle}{12} m_Q \frac{k^2 + m_Q}{(k^2 - m_Q^2)^4}. \quad (20)$$

Equivalently,

$$g_s G_A^{\alpha\beta} T^A S_{Q,\alpha\beta}^{(G)}(k) = -\frac{g_s G_A^{\alpha\beta} T^A}{4} \frac{\sigma_{\alpha\beta}(\not{k} + m_Q) + (\not{k} + m_Q)\sigma_{\alpha\beta}}{(k^2 - m_Q^2)^2}. \quad (21)$$

The one-gluon fields are contracted with

$$\langle g_s^2 G_A^{\alpha\beta} G_B^{\rho\sigma} \rangle = \frac{\delta_{AB}}{96} \langle g_s^2 G^2 \rangle \left(g^{\alpha\rho} g^{\beta\sigma} - g^{\alpha\sigma} g^{\beta\rho} \right), \quad (22)$$

and

$$\text{Tr}(T^A T^B) = \frac{\delta^{AB}}{2}. \quad (23)$$

Thus the color factor for the $S^{(G)}S^{(G)}$ piece is

$$\frac{N_c^2 - 1}{192} \langle g_s^2 G^2 \rangle, \quad N_c = 3. \quad (24)$$

The three projected OPE contributions are generated from

$$\Pi_1^{AA} \sim P_{(1)}^{\mu\nu} \text{Tr} [\gamma_\mu \gamma_5 S_c(k) \gamma_\nu \gamma_5 S_b(k-p)], \quad (25)$$

$$\Pi_1^{AB} \sim P_{(1)}^{\mu\nu} \text{Tr} \left[\gamma_\mu \gamma_5 S_c(k) i\sigma_{\nu\alpha} \frac{p^\alpha}{m_b + m_c} \gamma_5 S_b(k-p) \right], \quad (26)$$

$$\Pi_1^{BB} \sim P_{(1)}^{\mu\nu} \text{Tr} \left[i\sigma_{\mu\alpha} \frac{p^\alpha}{m_b + m_c} \gamma_5 S_c(k) i\sigma_{\nu\beta} \frac{p^\beta}{m_b + m_c} \gamma_5 S_b(k-p) \right]. \quad (27)$$

7 Perturbative spectral densities

For the perturbative loop, impose the two-particle cut

$$k^2 = m_c^2, \quad (k-p)^2 = m_b^2, \quad (28)$$

which gives

$$k \cdot p = \frac{s + m_c^2 - m_b^2}{2}, \quad s = p^2. \quad (29)$$

Define

$$\lambda(s, m_b, m_c) = s^2 + m_b^4 + m_c^4 - 2sm_b^2 - 2sm_c^2 - 2m_b^2 m_c^2. \quad (30)$$

The perturbative spectral densities are implemented as

$$\rho_{\text{pert}}^{ij}(s) = \frac{N_c}{16\pi^2} \frac{\sqrt{\lambda(s, m_b, m_c)}}{s} \frac{N^{ij}(s)}{(m_b + m_c)^{n_B^{ij}}}, \quad (31)$$

where $n_B^{AA} = 0$, $n_B^{AB} = n_B^{BA} = 1$, and $n_B^{BB} = 2$ counts how many tensor currents occur in the channel. The unnormalized numerator functions are

$$N^{AA}(s) = -\frac{2}{s} [(m_b + m_c)^2 - s] [(m_b - m_c)^2 + 2s], \quad (32)$$

$$N^{AB}(s) = 6(m_b - m_c) [(m_b + m_c)^2 - s], \quad (33)$$

$$N^{BB}(s) = 2 [-2(m_b^2 - m_c^2)^2 + (m_b^2 - 6m_b m_c + m_c^2)s + s^2]. \quad (34)$$

Equivalently, after setting $N_c = 3$,

$$\rho_{\text{pert}}^{AA}(s) = -\frac{3}{8\pi^2 s^2} [(m_b + m_c)^2 - s] [(m_b - m_c)^2 + 2s] \sqrt{\lambda(s, m_b, m_c)}, \quad (35)$$

$$\rho_{\text{pert}}^{AB}(s) = \frac{9}{8\pi^2 s(m_b + m_c)} (m_b - m_c) [(m_b + m_c)^2 - s] \sqrt{\lambda(s, m_b, m_c)}, \quad (36)$$

$$\rho_{\text{pert}}^{BB}(s) = \frac{3}{8\pi^2 s(m_b + m_c)^2} [-2(m_b^2 - m_c^2)^2 + (m_b^2 - 6m_b m_c + m_c^2)s + s^2] \sqrt{\lambda(s, m_b, m_c)}. \quad (37)$$

The perturbative Borel moments are

$$\Pi_{\text{pert}}^{ij}(M^2, s_0) = \int_{(m_b + m_c)^2}^{s_0} ds e^{-s/M^2} \rho_{\text{pert}}^{ij}(s). \quad (38)$$

With this convention all three perturbative spectral densities have mass dimension 2. Without the factors of $(m_b + m_c)^{-n_B^{ij}}$, their dimensions would be 2, 3, and 4 for AA , AB , and BB , respectively.

8 Dimension-4 and dimension-6 gluon condensates

The implemented condensate contribution is

$$\Pi_{G^2}^{ij} = \Pi^{ij}[S_c^{(G^2)} S_b^{(0)}] + \Pi^{ij}[S_c^{(0)} S_b^{(G^2)}] + \Pi^{ij}[S_c^{(G)} S_b^{(G)}]. \quad (39)$$

After Feynman parametrization the terms have the schematic form

$$\mathcal{A}_{G^2}^{ij}(x_1, x_2, s) = \sum_r C_r^{ij}(x_1, x_2, s) [m_c^2 x_1^2 + (m_b^2 + m_c^2 - s)x_1 x_2 + m_b^2 x_2^2]^{-n_r}. \quad (40)$$

The code then fixes the simplex by

$$x_1 = x, \quad x_2 = 1 - x. \quad (41)$$

This gives

$$\Delta(x, s) = m_c^2 x + m_b^2 (1 - x) - s x (1 - x). \quad (42)$$

For the triple-gluon condensate we use the standard momentum-space vacuum-averaged heavy-quark propagator term

$$S_Q^{(G^3)}(k) = \frac{\langle g_s^3 G^3 \rangle}{48} \frac{(\not{k} + m_Q) [\not{k}(k^2 - 3m_Q^2) + 2m_Q(2k^2 - m_Q^2)] (\not{k} + m_Q)}{(k^2 - m_Q^2)^6}. \quad (43)$$

The implemented dimension-6 contribution is

$$\Pi_{G^3}^{ij} = \Pi^{ij}[S_c^{(G^3)} S_b^{(0)}] + \Pi^{ij}[S_c^{(0)} S_b^{(G^3)}]. \quad (44)$$

This is the standard single-heavy-propagator contribution. It does not yet include any possible cross-line open-field terms of the schematic type $S^{(G_{\text{open}}^2)} S^{(G)}$, which require an explicitly open two-gluon propagator and a separate color/Lorentz derivation.

The G^2 and G^3 terms are not treated as ordinary smooth spectral densities. We instead use a direct Borel transform. Set $s = -Q^2$ and write

$$\Delta(x, -Q^2) = x(1-x) [Q^2 + \bar{s}(x)], \quad (45)$$

where

$$\bar{s}(x) = \frac{m_c^2 x + m_b^2 (1-x)}{x(1-x)}. \quad (46)$$

The Feynman-parameter amplitude is decomposed as

$$-i \mathcal{A}_{Gr}^{ij}(x, -Q^2) = \sum_{\ell \geq 1} \frac{a_\ell^{ij}(x)}{[Q^2 + \bar{s}(x)]^\ell} + \text{polynomial subtraction terms}. \quad (47)$$

The polynomial subtraction terms vanish after the Borel transform. The working phase convention is the explicit factor $-i$ shown above. This is a bookkeeping convention for removing the loop-integration prefactor returned by FeynCalc's Feynman-parameter amplitude. This is the convention that should be independently checked.

We use

$$\mathcal{B}_{M^2} \left[\frac{1}{(Q^2 + \bar{s})^n} \right] = \frac{1}{(n-1)! (M^2)^{n-1}} e^{-\bar{s}/M^2}. \quad (48)$$

Thus

$$\Pi_{Gr}^{ij}(M^2, s_0) = \int_{x_-}^{x_+} dx e^{-\bar{s}(x)/M^2} \sum_{\ell \geq 1} \frac{a_\ell^{ij}(x)}{(\ell-1)! (M^2)^{\ell-1}}. \quad (49)$$

The continuum restriction is

$$\bar{s}(x) \leq s_0, \quad (50)$$

which gives

$$x_\pm = \frac{s_0 + m_b^2 - m_c^2 \pm \sqrt{\lambda(s_0, m_b, m_c)}}{2s_0}. \quad (51)$$

9 Total Borel moments and numerical test point

The total moments used in the current numerical result are

$$\Pi^{ij}(M^2, s_0) = \Pi_{\text{pert}}^{ij}(M^2, s_0) + \Pi_{G^2}^{ij}(M^2, s_0) + \Pi_{G^3}^{ij}(M^2, s_0). \quad (52)$$

The current default numerical inputs are

$$m_b = 4.18 \text{ GeV}, \quad (53)$$

$$m_c = 1.27 \text{ GeV}, \quad (54)$$

$$\langle g_s^2 G^2 \rangle = 4\pi^2 (0.012) \text{ GeV}^4, \quad (55)$$

$$\langle g_s^3 G^3 \rangle = 0.57 \text{ GeV}^6, \quad (56)$$

$$M^2 = 10 \text{ GeV}^2, \quad (57)$$

$$s_0 = 55 \text{ GeV}^2. \quad (58)$$

For this point,

$$x_- = 0.3322728664, \quad x_+ = 0.9560816791. \quad (59)$$

The perturbative moments are

$$\Pi_{\text{pert}}^{AA} = 0.1441716826, \quad (60)$$

$$\Pi_{\text{pert}}^{AB} = -0.1051577192, \quad (61)$$

$$\Pi_{\text{pert}}^{BB} = 0.1348134513. \quad (62)$$

The gluon-condensate moments are

$$\Pi_{G^2}^{AA} = -0.0003658277, \quad (63)$$

$$\Pi_{G^2}^{AB} = 0.0002309420, \quad (64)$$

$$\Pi_{G^2}^{BB} = -0.0001888869. \quad (65)$$

The triple-gluon-condensate moments are

$$\Pi_{G^3}^{AA} = 0.0000700979, \quad (66)$$

$$\Pi_{G^3}^{AB} = -0.0000493659, \quad (67)$$

$$\Pi_{G^3}^{BB} = 0.0000310238. \quad (68)$$

Thus

$$\Pi_{\text{total}}^{AA} = 0.1438759528, \quad (69)$$

$$\Pi_{\text{total}}^{AB} = -0.1049761432, \quad (70)$$

$$\Pi_{\text{total}}^{BB} = 0.1346555882. \quad (71)$$

The corresponding angles are

$$\theta_{\text{pert}} = 43.726118943^\circ, \quad (72)$$

$$\theta_{\text{pert}+G^2} = 43.747426660^\circ, \quad (73)$$

$$\theta_{\text{pert}+G^2+G^3} = 43.742693524^\circ. \quad (74)$$

At this point the relative condensate corrections are

$$\frac{\Pi_{G^2}^{AA}}{\Pi_{\text{pert}}^{AA}} = -0.002537, \quad (75)$$

$$\frac{\Pi_{G^2}^{AB}}{\Pi_{\text{pert}}^{AB}} = -0.002196, \quad (76)$$

$$\frac{\Pi_{G^2}^{BB}}{\Pi_{\text{pert}}^{BB}} = -0.001401, \quad (77)$$

$$\frac{\Pi_{G^3}^{AA}}{\Pi_{\text{pert}}^{AA}} = 0.000486, \quad (78)$$

$$\frac{\Pi_{G^3}^{AB}}{\Pi_{\text{pert}}^{AB}} = 0.000469, \quad (79)$$

$$\frac{\Pi_{G^3}^{BB}}{\Pi_{\text{pert}}^{BB}} = 0.000230. \quad (80)$$

10 Perturbative α_s sensitivity model

The full perturbative $O(\alpha_s)$ correction is not yet calculated. As a diagnostic, the code implements a channel-dependent trial rescaling of the leading perturbative moment,

$$\Pi_{\text{pert}}^{ij}(M^2, s_0) \longrightarrow \left[1 + \frac{\alpha_s}{\pi} K_{ij}\right] \Pi_{\text{pert}}^{ij}(M^2, s_0). \quad (81)$$

The condensate pieces are left unchanged, so for the current total moment

$$\Pi_K^{ij} = \left[1 + \frac{\alpha_s}{\pi} K_{ij}\right] \Pi_{\text{pert}}^{ij} + \Pi_{G^2}^{ij} + \Pi_{G^3}^{ij}. \quad (82)$$

This is only a sensitivity model; it is not a substitute for the real two-loop NLO perturbative spectral density.

For example, with $\alpha_s = 0.26$ and

$$K_{AA} = 1, \quad K_{AB} = 0, \quad K_{BB} = -1, \quad (83)$$

the total angle at $M^2 = 10 \text{ GeV}^2$, $s_0 = 55 \text{ GeV}^2$ changes from

$$\theta_{\text{base}} = 43.742693524^\circ \quad (84)$$

to

$$\theta_K = 40.625721673^\circ, \quad \Delta\theta = -3.116971851^\circ. \quad (85)$$

A coarse scan with $K_{AA}, K_{AB}, K_{BB} \in \{-2, 0, 2\}$ gives a maximum branch-corrected shift of about

$$\max |\Delta\theta| \simeq 7.52^\circ. \quad (86)$$

11 Monte Carlo uncertainty propagation

The numerical input table gives central values,

$$m_b = 4.18 \text{ GeV}, \quad m_c = 1.27 \text{ GeV}, \quad (87)$$

$$\langle g_s^2 G^2 \rangle = 4\pi^2 (0.012) \text{ GeV}^4, \quad \langle g_s^3 G^3 \rangle = 0.57 \text{ GeV}^6. \quad (88)$$

For the final quoted error these inputs should be assigned allowed intervals or probability distributions. In the present implementation we use uniform sampling over editable intervals. A Monte Carlo point is

$$X_i = \left\{ m_b^{(i)}, m_c^{(i)}, G_2^{(i)}, G_3^{(i)}, M_i^2, s_{0,i} \right\}, \quad (89)$$

where

$$G_2 \equiv \langle g_s^2 G^2 \rangle, \quad G_3 \equiv \langle g_s^3 G^3 \rangle. \quad (90)$$

Each component is drawn from the user-specified interval

$$x^{(i)} \sim U(x_{\min}, x_{\max}). \quad (91)$$

Fixed parameters are implemented by setting $x_{\min} = x_{\max}$, or in the Mathematica code by passing a single number instead of an interval.

The current working ranges in the code are

$$m_b \in [4.16, 4.21] \text{ GeV}, \quad m_c \in [1.25, 1.29] \text{ GeV}, \quad (92)$$

$$G_2 \in 4\pi^2 [0.006, 0.018] \text{ GeV}^4, \quad G_3 \in [0.28, 0.86] \text{ GeV}^6, \quad (93)$$

$$M^2 \in [8, 12] \text{ GeV}^2, \quad s_0 \in [50, 60] \text{ GeV}^2. \quad (94)$$

Equivalently, in the central-value-plus-error notation useful for the paper,

$$m_b = (4.185 \pm 0.025) \text{ GeV}, \quad m_c = (1.27 \pm 0.02) \text{ GeV}, \quad (95)$$

$$G_2 = 4\pi^2(0.012 \pm 0.006) \text{ GeV}^4, \quad G_3 = (0.57 \pm 0.29) \text{ GeV}^6, \quad (96)$$

$$M^2 = (10 \pm 2) \text{ GeV}^2, \quad s_0 = (55 \pm 5) \text{ GeV}^2. \quad (97)$$

These are not final physics choices. They are placeholders for the scan layer and should be replaced by the intervals adopted in the analysis.

For every random point we impose the physical continuum condition

$$s_{0,i} > \left(m_b^{(i)} + m_c^{(i)} \right)^2. \quad (98)$$

Accepted points are evaluated with the same OPE formula as above,

$$\theta_i = \frac{1}{2} \text{ArcTan} \left[\Pi_1^{AA}(X_i) - \Pi_1^{BB}(X_i), -2\Pi_1^{AB}(X_i) \right], \quad (99)$$

using the chosen OPE order, normally “total” for $\Pi_{\text{pert}} + \Pi_{G^2} + \Pi_{G^3}$. For the present fast uncertainty scan we keep the triple-gluon term switched off by default. In the Mathematica implementation this is controlled by `$BcMixingMonteCarloIncludeG3=False` or by the option `"IncludeG3" -> False`. With this switch, a requested “total” Monte Carlo scan is evaluated as “pertG2”,

$$\Pi_{\text{MC}}^{ij} = \Pi_{\text{pert}}^{ij} + \Pi_{G^2}^{ij}, \quad (100)$$

and the G^3 contribution can be restored later by setting `"IncludeG3" -> True`.

The central value and Gaussian-width estimate are then

$$\bar{\theta} = \frac{1}{N} \sum_{i=1}^N \theta_i, \quad \sigma_{\theta, \text{Gauss}} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\theta_i - \bar{\theta})^2}. \quad (101)$$

The code also reports the unbiased sample standard deviation, median, minimum, maximum, and the 16% and 84% quantiles. These quantiles are useful because the resulting θ distribution may not be exactly Gaussian.

The Mathematica entry point is

```
nSamples = 500;
histBins = 25;
m2Range = {8.0, 12.0};
s0Range = {50.0, 60.0};

ranges = Join[
  $BcMixingDefaultUncertaintyRanges,
  <|"M2" -> m2Range, "s0" -> s0Range|>
];

uncertaintyRun = MonteCarloMixingAngleUncertainty[
  nSamples, ranges, "total",
  "IncludeG3" -> False,
  "Seed" -> 1234,
  "Progress" -> True
];
uncertaintyRun["Summary"]
MonteCarloMixingAngleHistogram[uncertaintyRun, histBins]
MonteCarloMixingAnglePublicationHistogram[uncertaintyRun, histBins]
```

where `ranges` is an association such as

```
<|
  "mb" -> {4.16, 4.21},
  "mc" -> {1.25, 1.29},
  "G2" -> {4 Pi^2 0.006, 4 Pi^2 0.018},
  "G3" -> {0.28, 0.86},
  "M2" -> {8.0, 12.0},
  "s0" -> {50.0, 60.0}
|>
```

The number of random points, the input intervals, and the Borel window are therefore all editable without touching the algebra.

The first argument, for example 500, is the number of random accepted Monte Carlo points. The option `"Seed" -> 1234` fixes the random-number seed so the sample is reproducible. The Monte Carlo result is called `uncertaintyRun`, not `mc`, to avoid confusion with the charm mass parameter `"mc"`. To vary only the Borel window and the continuum threshold while keeping the QCD inputs fixed at their central values, one can use

```
ranges = Join[
  $BcMixingDefaultUncertaintyRanges,
  <|
    "mb" -> 4.18,
    "mc" -> 1.27,
    "G2" -> 4 Pi^2 0.012,
    "G3" -> 0.57,
    "M2" -> {8.0, 12.0},
    "s0" -> {50.0, 60.0}
  |>
];
```

The accepted random points and the summary can be exported for replotting:

```
ExportMonteCarloMixingAngleSamples[uncertaintyRun, "BcMixingMonteCarloSamples.csv"];
ExportMonteCarloMixingAngleSummary[uncertaintyRun, "BcMixingMonteCarloSummary.csv"];
Export["BcMixingMonteCarloHistogram.pdf",
  MonteCarloMixingAnglePublicationHistogram[uncertaintyRun, 25]];
```

12 How to reproduce the numerical check

In Mathematica, load the implementation with

```
SetDirectory["/Users/sbilmis/Bc_mixing"];
Get["BcMixingMomentum.wl"];
```

Then evaluate

```
NumericOPESummary[10, 55]
NumericMixingAngleDegrees[10, 55, "pert"]
NumericMixingAngleDegrees[10, 55, "pertG2"]
NumericMixingAngleDegrees[10, 55, "total"]
KFactorSensitivityRecord[10, 55, KFactorAssociation[1, 0, -1], 0.26, "total"]
KFactorSensitivityEnvelope[10, 55, {-2, 2, 2}, 0.26, "total"]
MixingMassDimensionReport[]
CheckMixingMassDimensions[]
PerturbativeSpectralDensityDimensionReport[]
```

For window studies one can scan

```
MixingAngleTable[{8, 12, 1}, {50, 60, 5}, "total"]
OPEConvergenceDataset[{8, 12, 1}, {50, 60, 5}]
uncertaintyRun = MonteCarloMixingAngleUncertainty[50, $BcMixingDefaultUncertaintyRanges,
  "total", "IncludeG3" -> False, "Seed" -> 1234]
uncertaintyRun["Summary"]
```

13 Auxiliary-parameter stability plots

As a comparison with the vector B_c^* analysis of Wang, we also consider the auxiliary window

$$M^2 \in [7.0, 9.0] \text{ GeV}^2, \quad s_0 = (54 \pm 1) \text{ GeV}^2. \quad (102)$$

The Mathematica implementation stores this as

```
$BcMixingWangWindow
```

and produces publication-style stability plots with

```
stabilityOrder = "pertG2";
```

```
m2Stability = MixingAngleStabilityM2PublicationPlot[
  $BcMixingWangWindow["M2Range"],
  $BcMixingWangWindow["s0Values"],
  stabilityOrder,
  $BcMixingDefaultParameters,
  "NPoints" -> 25,
  "YHalfWidth" -> 1.0
];
```

```
s0Stability = MixingAngleStabilityS0PublicationPlot[
  $BcMixingWangWindow["s0Range"],
  $BcMixingWangWindow["M2Values"],
  stabilityOrder,
  $BcMixingDefaultParameters,
  "NPoints" -> 25,
  "YHalfWidth" -> 1.0
];
```

The option "YHalfWidth" -> 1.0 prevents the vertical axis from being over-zoomed; it enforces at least a $\pm 1^\circ$ display window around the calculated angles. This is useful when the goal is to demonstrate stability with respect to the auxiliary parameters, not to magnify tiny numerical variations. For publication labels the plotting functions try to use the Mathematica package MaTeX automatically,

```
"UseMaTeX" -> Automatic
```

so that the axes and legends are rendered as $M^2(\text{GeV}^2)$, $s_0(\text{GeV}^2)$, θ° , and entries such as $s_0 = 54 \text{ GeV}^2$. If MaTeX is not installed, the code falls back to ordinary Mathematica labels with a degree symbol. The fallback can be forced with "UseMaTeX" -> False.

The plots and their underlying data can be exported with

```

Export["BcMixingThetaVsM2_WangWindow.pdf", m2Stability["Plot"]];
Export["BcMixingThetaVsS0_WangWindow.pdf", s0Stability["Plot"]];
ExportStabilityDataCSV[m2Stability["Data"],
  "BcMixingThetaVsM2_WangWindow.csv", "M2"];
ExportStabilityDataCSV[s0Stability["Data"],
  "BcMixingThetaVsS0_WangWindow.csv", "S0"];

```

14 Coordinate-space cross-check

The file `BcMixingCoordinate.wl` starts an independent coordinate-space validation of the perturbative part. This is meant as a blind check of the momentum-space calculation, not as a replacement of it. The same interpolating currents and the same normalization of the tensor current are used,

$$J_\mu^A(x) = \bar{b}(x)\gamma_\mu\gamma_5 c(x), \quad J_\mu^B(x) = i\bar{b}(x)\sigma_{\mu\alpha}\frac{p^\alpha}{m_b + m_c}\gamma_5 c(x). \quad (103)$$

With these conventions, the coordinate-space perturbative correlators are

$$\Pi_{\mu\nu}^{AA}(p) = -iN_c \int d^4x e^{ip\cdot x} \text{Tr} \left[\gamma_\mu \gamma_5 S_c^{(0)}(-x) \gamma_\nu \gamma_5 S_b^{(0)}(x) \right], \quad (104)$$

$$\Pi_{\mu\nu}^{AB}(p) = -iN_c \int d^4x e^{ip\cdot x} \text{Tr} \left[\gamma_\mu \gamma_5 S_c^{(0)}(-x) i\sigma_{\nu\beta} \frac{p^\beta}{m_b + m_c} \gamma_5 S_b^{(0)}(x) \right], \quad (105)$$

$$\Pi_{\mu\nu}^{BB}(p) = -iN_c \int d^4x e^{ip\cdot x} \text{Tr} \left[i\sigma_{\mu\alpha} \frac{p^\alpha}{m_b + m_c} \gamma_5 S_c^{(0)}(-x) i\sigma_{\nu\beta} \frac{p^\beta}{m_b + m_c} \gamma_5 S_b^{(0)}(x) \right]. \quad (106)$$

The sign difference between $S_c(-x)$ and $S_b(x)$ is implemented in the code by changing the sign of the $\not{x}$ term. The spin-1 invariant amplitude is extracted with the same transverse projector as in momentum space,

$$\Pi_1^{ij}(p^2) = \frac{1}{3} \left(g^{\mu\nu} - \frac{p^\mu p^\nu}{p^2} \right) \Pi_{\mu\nu}^{ij}(p). \quad (107)$$

This step is performed in `CoordinateTraceKernel[channel,"pert"]`.

The free heavy-quark propagator is written in terms of modified Bessel functions,

$$S_Q^{(0)}(x) = \frac{m_Q^2}{(2\pi)^2} \left[i\not{x} \frac{K_2(m_Q\sqrt{-x^2})}{-x^2} + \frac{K_1(m_Q\sqrt{-x^2})}{\sqrt{-x^2}} \right]. \quad (108)$$

This is the form used in the coordinate-space treatments of heavy systems. The derivative-free perturbative trace contains products of the form

$$(\not{x})^a (p \cdot x)^b \frac{K_\nu(m_c\sqrt{-x^2}) K_\mu(m_b\sqrt{-x^2})}{(\sqrt{-x^2})^n}, \quad (109)$$

where the tensor numerator is reduced by `FeynCalc` before the radial Fourier integral is performed.

After Wick rotation to Euclidean momentum $Q^2 = -p^2$, the angular integral has the standard form

$$i \int d^Dx e^{ip\cdot x} f(\sqrt{-x^2}) = (2\pi)^{D/2} Q^{1-D/2} \int_0^\infty dx x^{D/2} J_{D/2-1}(Qx) f(x). \quad (110)$$

Therefore every scalar term is reduced to radial integrals

$$I_{\nu\nu\mu}(Q; m_c, m_b) = \int_0^\infty dx x^{u-1} J_\nu(Qx) K_\nu(m_c x) K_\mu(m_b x). \quad (111)$$

For equal heavy masses, Huang–Liu give a compact hypergeometric expression for this integral. For two different heavy masses, the Azizi et al. Bessel/Schwinger representation leads to the same physical two-body cut. The installed perturbative spectral density is

$$\rho_{\text{pert}}^{ij}(s) = \frac{N_c}{16\pi^2} \frac{\sqrt{\lambda(s, m_b, m_c)}}{s} N^{ij}(s) (m_b + m_c)^{-n_{ij}}, \quad (112)$$

where $n_{AA} = 0$, $n_{AB} = 1$, $n_{BB} = 2$, and

$$\lambda(s, m_b, m_c) = s^2 + m_b^4 + m_c^4 - 2sm_b^2 - 2sm_c^2 - 2m_b^2m_c^2, \quad (113)$$

$$k \cdot p|_{\text{cut}} = \frac{s + m_c^2 - m_b^2}{2}. \quad (114)$$

The numerator functions are

$$N^{AA}(s) = -\frac{4}{s} [2(k \cdot p)^2 + (3m_b m_c + m_c^2)s - 3s(k \cdot p)], \quad (115)$$

$$N^{AB}(s) = 12 [-(m_b + m_c)(k \cdot p) + m_c s], \quad (116)$$

$$N^{BB}(s) = -4 [4(k \cdot p)^2 + (3m_b m_c - m_c^2)s - 3s(k \cdot p)]. \quad (117)$$

A hand check should verify these three numerators from the projected coordinate-space trace and then verify the phase-space factor $N_c \sqrt{\lambda}/(16\pi^2 s)$. In the code these expressions are stored in `CoordinatePerturbativeNumerator` and `CoordinatePerturbativeSpectralDensity`.

The perturbative Borel moment is then evaluated as

$$\Pi_{\text{pert}}^{ij}(M^2, s_0) = \int_{(m_b+m_c)^2}^{s_0} ds e^{-s/M^2} \rho_{\text{pert}}^{ij}(s), \quad (118)$$

and the angle is obtained from the same diagonalization formula,

$$\theta_{\text{coord}}^{\text{pert}}(M^2, s_0) = \frac{1}{2} \text{ArcTan} [\Pi_{\text{pert}}^{AA} - \Pi_{\text{pert}}^{BB}, -2\Pi_{\text{pert}}^{AB}]. \quad (119)$$

14.1 What should be written in the journal paper

If the paper presents the coordinate-space calculation as the main method, the main text should not merely state that the result was obtained with Mathematica. It should show enough of the analytic reduction that a reader can see where the physical cut and continuum subtraction enter. A compact but sufficient presentation would include:

1. the two currents, including the tensor-current normalization $p^\alpha/(m_b + m_c)$;
2. the coordinate-space correlators after Wick contraction;
3. the heavy-quark propagator in terms of $K_\nu(m_Q \sqrt{-x^2})$;
4. the spin-1 projection used to define Π_1^{ij} ;
5. the generic Bessel integral generated by the Fourier transform;
6. the way the Bessel functions are converted to Schwinger-parameter integrals;
7. the emergence of Dirac delta functions after the Borel transform;
8. the resulting Heaviside support conditions or, equivalently, the integration limits that implement continuum subtraction;
9. the final perturbative spectral densities, preferably in the compact $N^{ij}(s)\sqrt{\lambda}/s$ form;

10. the statement that the current coordinate-space condensate check includes local G^2 and G^3 single-line terms, while the full dimension-4 coordinate-space result additionally needs the $S_c^{(G)} S_b^{(G)}$ cross-line term.

The main text can contain the compact formulas and the final spectral densities. Longer channel-by-channel trace expressions and the local condensate Borel kernels are better placed in an appendix or supplemental file.

14.2 Role of the Azizi et al. method

The main technical role of the Azizi et al. method is to make continuum subtraction well defined for systems with more than one heavy propagator in coordinate space. With one heavy quark, the coordinate-space Fourier transform contains one modified Bessel function and can often be reduced to ordinary hypergeometric functions. With two heavy quarks, the Fourier transform contains products

$$J_\nu(Qx)K_\nu(m_c x)K_\mu(m_b x), \quad (120)$$

and a naive treatment of the Bessel functions can lead to parameter integrals with apparent endpoint divergences or ambiguous continuum subtraction.

Azizi et al. avoid this by using an integral representation of the modified Bessel functions and then applying the Borel transformations before doing all parameter integrations. In this order of operations, the Borel transforms generate Dirac delta functions. These delta functions fix combinations of Schwinger parameters and convert the would-be divergent parameter domains into finite domains with explicit support conditions. The support conditions are usually written as Heaviside functions or, after solving the inequalities, as modified integration limits.

Without this organization one would have to handle the delta functions, derivatives of delta functions, endpoint constraints, and continuum subtraction analytically by hand for each term. That is possible, but it is error-prone for two heavy masses because every numerator power changes the derivative order of the delta function or the power of the support factor. The Azizi formulation gives a systematic recipe:

$$\begin{aligned} K_\nu K_\mu &\longrightarrow \text{Schwinger parameters} \longrightarrow \text{Borel delta functions} \\ &\longrightarrow \Theta\text{-functions} \longrightarrow \text{finite integration limits.} \end{aligned} \quad (121)$$

In the paper, this chain should be shown once for a generic term. The final channel-dependent spectral densities can then be quoted in compact form.

14.3 Direct coordinate-space route without the Azizi reduction

It is useful to state explicitly what would happen if one deliberately did not use the Azizi et al. organization. This is not the route used by the current numerical code, and it should be treated as an independent analytic cross-check or appendix-level exercise. The starting point is still the same coordinate-space correlator,

$$\Pi_{\mu\nu}^{ij}(p) = -iN_c \int d^4x e^{ip \cdot x} \text{Tr} [\Gamma_\mu^i S_c(-x) \Gamma_\nu^j S_b(x)], \quad (122)$$

but after the Dirac trace one keeps the Fourier transform of each Bessel product directly. A typical scalar integral has the form

$$\mathcal{I}_{n\mu\nu}^{(r)}(p^2) = \int d^4x e^{ip \cdot x} (p \cdot x)^r (x^2)^n K_\mu(m_c \sqrt{-x^2}) K_\nu(m_b \sqrt{-x^2}). \quad (123)$$

The powers of $p \cdot x$ may be generated by differentiating the scalar Fourier integral with respect to p_α ,

$$x_\alpha e^{ip \cdot x} = \frac{1}{i} \frac{\partial}{\partial p^\alpha} e^{ip \cdot x}, \quad (124)$$

or by doing Lorentz tensor reduction before the radial integral. The angular integration then gives integrals containing

$$J_1(Qx)K_\mu(m_c x)K_\nu(m_b x), \quad Q^2 = -p^2. \quad (125)$$

At this stage the direct method must extract the discontinuity in $p^2 = s$ from the Bessel product itself. If Schwinger parameters are introduced only as a calculational aid, one meets distributions of the type

$$\delta\left(s - \frac{m_c^2}{x} - \frac{m_b^2}{1-x}\right), \quad (126)$$

and, when numerator powers or derivatives act on the Fourier kernel, derivatives of the same delta function,

$$\delta^{(n)}(s - \bar{s}(x)), \quad \bar{s}(x) = \frac{m_c^2}{x} + \frac{m_b^2}{1-x}. \quad (127)$$

These objects must be treated analytically. For example,

$$\int_0^1 dx F(x) \delta(s - \bar{s}(x)) = \sum_{x_i} \frac{F(x_i)}{|\bar{s}'(x_i)|}, \quad \bar{s}(x_i) = s, \quad (128)$$

where the roots x_i exist only if $s \geq (m_b + m_c)^2$. This condition is the same physical threshold and may be written as a Heaviside factor,

$$\Theta(s - (m_b + m_c)^2). \quad (129)$$

For derivative delta functions one must integrate by parts,

$$\int dx F(x) \delta^{(n)}(s - \bar{s}(x)), \quad (130)$$

carefully keeping the Jacobian factors and verifying that no endpoint terms survive after the physical support restriction is imposed.

Thus the direct route has the schematic structure

$$\begin{aligned} \text{Bessel product} &\longrightarrow \text{radial Fourier transform} \longrightarrow \text{Disc}_{p^2=s} \\ &\longrightarrow \delta(s - \bar{s}), \delta'(s - \bar{s}), \dots \longrightarrow \Theta(s - s_{\text{th}}) \longrightarrow x_{\pm}(s). \end{aligned} \quad (131)$$

This is analytically possible, but it is precisely the bookkeeping that the Azizi et al. method systematizes. The direct section is therefore best used as a validation path: derive one channel, preferably AA , all the way to the same perturbative spectral density and confirm that the final answer is proportional to

$$\frac{\sqrt{\lambda(s, m_b, m_c)}}{s} \Theta(s - (m_b + m_c)^2). \quad (132)$$

Only after the AA channel is reproduced should one repeat the calculation for AB and BB , because those channels additionally test the tensor current normalization $1/(m_b + m_c)$. The condensate terms are substantially more cumbersome in this route, since local G^2 , local G^3 , and the open-field $S_c^{(G)} S_b^{(G)}$ term can generate higher derivatives of the same support delta functions.

14.4 Where the Dirac delta and Heaviside functions enter

The reason the current Mathematica output does not visibly display δ -functions and Heaviside functions is not that they are absent from the coordinate-space method. Rather, in the current implementation they have already been integrated out or replaced by equivalent integration limits.

The coordinate-space product of two massive propagators generates scalar terms of the schematic form

$$T = \int d^4x e^{ip \cdot x} \frac{K_\nu(m_c \sqrt{-x^2}) K_\mu(m_b \sqrt{-x^2})}{(\sqrt{-x^2})^n} \times \{\text{polynomial in } x^2, p \cdot x, p^2\}. \quad (133)$$

Using an integral representation for each modified Bessel function, one obtains Schwinger-parameter integrals of the schematic form

$$T \sim \int d\rho dv dw \rho^a v^b w^c (1-v-w)^d \exp \left[-\frac{m_c^2}{\rho v} - \frac{m_b^2}{\rho w} - \frac{Q^2}{4\rho} \right], \quad Q^2 = -p^2. \quad (134)$$

This is the stage emphasized in the Azizi et al. treatment. In applications with two external virtualities, successive Borel transformations produce Dirac delta functions that fix combinations of the Schwinger parameters. A typical reduced structure is

$$\rho(s_1, s_2) \sim \int dz dy dl F(z, y, l) \delta \left[s_1 - l - \frac{m_c^2}{z(1-y)} - \frac{m_b^2}{zy} \right] \delta(s_2 - s_1), \quad (135)$$

up to derivatives of the delta functions when powers of the external invariants appear. After the l -integration, these delta functions turn into support restrictions,

$$\int_0^\infty dl F(l) \delta(s - l - A) = F(s - A) \Theta(s - A), \quad (136)$$

$$A = \frac{m_c^2}{z(1-y)} + \frac{m_b^2}{zy}. \quad (137)$$

Thus one obtains

$$\Theta \left(s_1 - \frac{m_c^2}{z(1-y)} - \frac{m_b^2}{zy} \right), \quad (138)$$

or equivalently into restricted integration limits. These Θ -functions are what remove apparent endpoint divergences in the Schwinger-parameter integrals.

For the present two-point mixing correlator there is one external invariant, $p^2 = s$, and the same physics appears in a simpler form: the spectral density has support only above

$$s_{\text{th}} = (m_b + m_c)^2, \quad (139)$$

or, in a Feynman-parameter representation,

$$s > \bar{s}(x) = \frac{m_c^2 x + m_b^2 (1-x)}{x(1-x)}. \quad (140)$$

Solving $\bar{s}(x) < s_0$ gives the finite integration interval

$$x_-(s_0) < x < x_+(s_0), \quad (141)$$

where

$$x_\pm(s_0) = \frac{s_0 + m_b^2 - m_c^2 \pm \sqrt{\lambda(s_0, m_b, m_c)}}{2s_0}. \quad (142)$$

This is the explicit replacement used in the numerical integrations:

$$\int_0^1 dx F(x) \Theta(s_0 - \bar{s}(x)) \longrightarrow \int_{x_-(s_0)}^{x_+(s_0)} dx F(x). \quad (143)$$

Equivalently, when writing the dispersion integral, the same support appears as

$$\int ds \rho(s) = \int ds \Theta(s - (m_b + m_c)^2) \rho(s), \quad (144)$$

with the continuum-subtracted Borel moment

$$\Pi(M^2, s_0) = \int_{(m_b+m_c)^2}^{s_0} ds e^{-s/M^2} \rho(s). \quad (145)$$

Thus the Heaviside support has not disappeared; it is encoded in the lower threshold and in the $x_{\pm}$ integration limits used by the numerical Borel integrals.

The present coordinate-space code has two layers:

1. For the perturbative check, it installs the final reduced spectral density proportional to $\sqrt{\lambda(s, m_b, m_c)} N^{ij}(s)/s$. Therefore no intermediate δ - or Θ -functions appear in the output.
2. For the local G^2 and G^3 checks, it uses the already-tested Feynman/Schwinger direct-Borel reduction helpers from `BcMixingMomentum.wl`. In this representation the support is again implemented by the integration limits $x_{\pm}(s_0)$, not by printing explicit Heaviside functions.

For a journal paper, it is therefore important to say explicitly that the Dirac delta functions and Heaviside functions occur at the Schwinger/Borel reduction stage, but in the numerical implementation they are already integrated out and appear as physical thresholds and finite parameter integration domains.

For a student checking this section by hand, the suggested coordinate-space exercise is:

1. Start from the AA current only. Insert $S_c^{(0)}(-x)$ and $S_b^{(0)}(x)$ and keep track of the sign of the $\not{x}$ term.
2. Evaluate the Dirac trace before projection. The trace should reduce to scalar combinations of x^2 , p^2 , and $p \cdot x$, multiplied by products of K_1 and K_2 .
3. Apply the transverse projector and replace $p^2 \rightarrow s$, $x^2 \rightarrow x^2$, and $p \cdot x \rightarrow px$. This is what `CoordinateScalarize` does.
4. Identify the radial integral associated with each scalar term. Terms with powers of $p \cdot x$ can be generated by differentiating the Fourier factor with respect to p_{μ} , or equivalently by using tensor reduction before the angular integral.
5. Use the Bessel/Schwinger representation to obtain the physical cut in s . The final perturbative answer must be proportional to $\sqrt{\lambda(s, m_b, m_c)}$ and vanish at $s = (m_b + m_c)^2$.
6. Repeat the numerator check for AB and BB . These two channels are the best tests of the tensor-current normalization because each B current contributes one factor $1/(m_b + m_c)$.

The coordinate-space notebook separates these checks intentionally:

- first evaluate only `BcMixingCoordinate.wl`;
- record `CoordinateTraceKernel["AA", "pert"]`;
- record `CoordinatePerturbativeSpectralDensity["AA", s]`;
- compute `CoordinateNumericMixingAngleDegrees[8, 54, "pert"]`;
- only after this blind check, load `BcMixingMomentum.wl` and compare with

`CoordinateCompareToMomentum[8, 54]`

The Mathematica entry points are

```

Get["BcMixingCoordinate.wl"];
CoordinateCheckEnvironment[]
CoordinateTraceKernel["AA", "pert"] // Short
CoordinateNumericMixingAngleDegrees[8, 54, "pert"]
CoordinateWangWindowGrid[]

```

If the momentum-space file is also loaded, the two perturbative calculations can be compared by

```

Get["BcMixingMomentum.wl"];
Get["BcMixingCoordinate.wl"];
CoordinateCompareToMomentum[8, 54]

```

which gives

```

<|"CoordinatePertDeg" -> 43.28944792785432,
  "MomentumPertDeg" -> 43.28944792785432,
  "DifferenceDeg" -> 0.|>

```

for $M^2 = 8 \text{ GeV}^2$ and $s_0 = 54 \text{ GeV}^2$.

For the next stage, the coordinate-space G^2 and G^3 propagator kernels are already written as extension points. The local-condensate heavy-quark terms used in the code are

$$S_Q^{(G^2)}(x) = -\frac{\langle g_s^2 G^2 \rangle}{2^6 3 (2\pi)^2} \left[(im_Q \not{x} - 6) \frac{K_1(m_Q \sqrt{-x^2})}{\sqrt{-x^2}} + 4m_Q x^2 \frac{K_2(m_Q \sqrt{-x^2})}{-x^2} \right], \quad (146)$$

$$S_Q^{(G^3)}(x) = \frac{\langle g_s^3 G^3 \rangle}{2^8 3^2 (2\pi)^2} \left[-\frac{i \not{x} x^2}{m_Q} \frac{K_1(m_Q \sqrt{-x^2})}{\sqrt{-x^2}} + 4i \not{x} x^2 \frac{K_2(m_Q \sqrt{-x^2})}{-x^2} \right. \\ \left. + \frac{10x^4}{m_Q} \frac{K_2(m_Q \sqrt{-x^2})}{-x^2} + x^4 \frac{K_1(m_Q \sqrt{-x^2})}{\sqrt{-x^2}} \right]. \quad (147)$$

The dimension-4 coordinate-space check must include

$$S_c^{(G^2)}(-x) S_b^{(0)}(x), \quad S_c^{(0)}(-x) S_b^{(G^2)}(x), \quad (148)$$

and, if the open-field propagator is kept explicitly, the cross-line $S_c^{(G)}(-x) S_b^{(G)}(x)$ term. The present coordinate file records the local G^2 and G^3 kernels and can build their projected trace kernels, for example

```

CoordinateTraceKernel["AA", "G2c"]      (* S_c^(G2) S_b^(0) *)
CoordinateTraceKernel["AA", "G2b"]      (* S_c^(0) S_b^(G2) *)
CoordinateTraceKernel["AA", "G2local"]  (* local sum of the two *)
CoordinateTraceKernel["AA", "G3local"]  (* local single-line G3 sum *)

```

At this stage, however, the numerical coordinate-space spectral/Borel integration is enabled directly only for "pert". A reduced Bessel/Schwinger integration bridge has now been added for the local single-line condensate pieces. It is used only after loading `BcMixingMomentum.wl`, because that file already contains the tested Feynman/Schwinger parameter reduction and direct Borel transform. The coordinate notebook therefore keeps these cells in an optional section:

```

Get["BcMixingMomentum.wl"];
Get["BcMixingCoordinate.wl"];

CoordinateLocalCondensateSummary[8, 54]

```

```

CoordinateNumericMixingAngleDegrees[8, 54, "pertG2local"]
CoordinateNumericMixingAngleDegrees[8, 54, "totalLocal"]
CoordinateLocalCondensateComparisonToMomentum[8, 54]
CoordinateLocalCondensateComparisonToMomentum[8, 54,
  $BcCoordinateDefaultParameters, "IncludeG3" -> True]

```

The labels mean

$$G2c : S_c^{(G^2)}(-x)S_b^{(0)}(x), \quad G2b : S_c^{(0)}(-x)S_b^{(G^2)}(x), \quad (149)$$

$$G3c : S_c^{(G^3)}(-x)S_b^{(0)}(x), \quad G3b : S_c^{(0)}(-x)S_b^{(G^3)}(x). \quad (150)$$

At $M^2 = 8 \text{ GeV}^2$, $s_0 = 54 \text{ GeV}^2$, the local-coordinate test gives

$$\theta_{\text{pert}+G2\text{local}} = 43.321950845^\circ, \quad \theta_{\text{pert}+G2\text{local}+G3\text{local}} = 43.308150036^\circ. \quad (151)$$

The word “local” is important. It means that the condensate is inserted on one heavy-quark line at a time. For G^3 , this reproduces the single-line momentum-space G^3 implementation used here. For G^2 , it does not yet reproduce the complete dimension-4 result, because there is also an open-field term in which one background gluon is attached to the charm line and one to the bottom line.

Numerically, for the AA channel at the same test point,

$$\Pi_{G^2,\text{local}}^{AA} = -3.1655383105 \times 10^{-4}, \quad (152)$$

$$\Pi_{G^2,\text{full momentum}}^{AA} = -3.6293326185 \times 10^{-4}, \quad (153)$$

$$\Pi_{G^2,\text{residual}}^{AA} = -4.6379430799 \times 10^{-5}. \quad (154)$$

This residual is the target for the missing coordinate-space $S_c^{(G)}(-x)S_b^{(G)}(x)$ derivation. By contrast, the local coordinate and momentum-space G^3 moments agree numerically at this point:

$$\Pi_{G^3,\text{local}}^{AA} = 6.8538572092 \times 10^{-5}, \quad (155)$$

$$\Pi_{G^3,\text{momentum}}^{AA} = 6.8538572092 \times 10^{-5}. \quad (156)$$

This should not yet be quoted as the complete condensate-corrected coordinate result. The reason is that the full dimension-4 contribution also contains the cross-line $S_c^{(G)}(-x)S_b^{(G)}(x)$ term. The helper `CoordinateLocalCondensateComparisonToMomentum` reports the difference between the full momentum-space G^2 moment and the local coordinate $G2c + G2b$ moment; this residual is the piece that must be reproduced from the open-field coordinate-space propagator. The G^3 comparison is optional because it is slower; setting `"IncludeG3" -> True` checks that the local coordinate $G3c + G3b$ integration matches the momentum-space G^3 contribution.

15 Checklist for independent verification

A theorist can check the calculation in the following order:

1. Confirm the current convention, especially the explicit factor of i in J_μ^B , the normalization $p^\alpha/(m_b + m_c)$, and the definition of $\sigma_{\mu\nu}$.
2. Reproduce the three projected perturbative traces and verify the numerator functions $N^{AA}(s)$, $N^{AB}(s)$, and $N^{BB}(s)$.
3. Check the two-body phase-space normalization $N_c\sqrt{\lambda}/(16\pi^2s)$ used in $\rho_{\text{pert}}^{ij}(s)$.

4. Verify the background-field propagator terms and the color factor in the $S^{(G)}S^{(G)}$ contribution.
5. Independently derive the direct Borel formula for the G^2 and G^3 terms, including the $-i$ phase convention.
6. Check whether the standard single-line G^3 propagator contribution is sufficient, or whether cross-line open-field G^3 terms must be added.
7. Check that $\Pi_{G^2}/\Pi_{\text{pert}}$ and $\Pi_{G^3}/\Pi_{\text{pert}}$ remain small in the accepted Borel window.
8. Use the K_{ij} sensitivity model only as an estimate of possible channel-dependent perturbative $O(\alpha_s)$ uncertainty.
9. Derive the true $O(\alpha_s)$ two-loop perturbative spectral densities separately.
10. Determine the final working window using OPE convergence, Borel stability, and pole-dominance criteria.